

```
In [1]: pip install pandas
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in C:\ProgramData\anaconda3\Lib\site-packages (3.0.3)
Requirement already satisfied: numpy>=2.3.3 in C:\ProgramData\anaconda3\Lib\site-packages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\ProgramData\anaconda3\Lib\site-packages (from pandas
) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\ProgramData\anaconda3\Lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: six>=1.5 in C:\ProgramData\anaconda3\Lib\site-packages (from python-dateutil>=2.8
.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
In [2]: import pandas as pd
data=pd.read_csv("C:/Users/RAUSHAN KUMAR/Downloads/trends.csv")
```

```
In [3]: data.head()
```

```
Out[3]:
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

```
In [4]: data.shape
```

```
Out[4]: (26955, 5)
```

```
In [5]: data.tail()
```

```
Out[5]:
```

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miền thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

```
In [6]: data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0    location    26955 non-null  str
1    year        26955 non-null  int64
2    category    26955 non-null  str
3    rank        26955 non-null  int64
4    query       26955 non-null  str
dtypes: int64(2), str(3)
memory usage: 2.0 MB
```

```
In [7]: data.describe()
```

```
Out[7]:
```

	year	rank
count	26955.000000	26955.000000
mean	2015.243369	3.000000
std	3.564683	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

```
In [9]: data.nunique()
```

```
Out[9]: location      83
        year         20
        category     2450
        rank          5
        query       18431
        dtype: int64
```

```
In [10]: data.isnull().sum()
```

```
Out[10]: location      0
        year         0
        category      0
        rank         0
        query        0
        dtype: int64
```

```
In [15]: int(data.duplicated().sum())
```

```
Out[15]: 0
```

```
In [16]: data.drop_duplicates(inplace=True)
```

```
In [18]: int(data.duplicated().sum())
```

```
Out[18]: 0
```

```
In [19]: data.info()
```

```
<class 'pandas.DataFrame'>
Index: 26945 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   location    26945 non-null  str
1   year        26945 non-null  int64
2   category    26945 non-null  str
3   rank        26945 non-null  int64
4   query       26945 non-null  str
dtypes: int64(2), str(3)
memory usage: 2.2 MB
```

```
In [ ]:
```